

A Model-Agnostic Feature Selection Technique to Improve the Performance of One-Class Classifiers

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Abstract—One-class classifiers hold promise for applications like fraudulent credit card transaction identification. However, interpreting these models to understand which features drive predictions is challenging. Such an understanding is necessary to avoid brute-force approaches to feature selection. This paper explores SHAP (SHapley Additive exPlanations) for feature selection with one-class classifiers on a credit card fraud dataset. We apply SHAP to select key features and evaluate One-Class Gaussian mixture models and One-Class Support Vector Machines. Statistical analysis tests show Gaussian mixture models built with SHAP-selected features perform significantly better than Gaussian mixture models built without feature selection. To the best of our knowledge, we are the first to show the benefit of SHAP-based feature selection to One-Class Gaussian mixture models. Moreover, we show that robust performance with features of the full dataset may be a prerequisite in order for SHAP feature selection to impart further gains. Our results provide novel evidence that SHAP can identify informative features for one-class classifiers.

Index Terms—One-class classifier, feature selection, imbalanced data, SHAP, Credit Card Fraud Detection

INTRODUCTION

In order for credit cards to be a reliable means of making payments, fraudulent credit card transactions must be kept to a minimum. Therefore, in a healthy credit card transaction ecosystem, the detection of fraudulent transactions is the capture of a rare event. An overabundance of fraudulent transactions would cause the system to collapse. Therefore, data on legitimate transactions is easier to find. The challenge of using purely legitimate transaction data in Machine Learning applications is that it is only of one class, so one may not employ Binary-Class classifiers. Hence, one may find an opportunity to employ One-Class classifiers such as One-Class Support Vector Machines (SVMs) [1] and One-Class Gaussian Mixture Models (GMMs) [2] as classifiers for Credit Card fraud detection. Hence, this is a study on the application of One-Class Classifiers (OCCs) for anomaly detection. For a study on unsupervised techniques for anomaly detection, please see [3].

We take such an opportunity here. To the best of our knowledge, we are the first to apply SHapley Additive exPlanations (SHAP) [4] in a feature selection technique for One-Class Classifiers (OCCs). Here, we provide a solution to the issue Bauder *et al.* demonstrate with diminishing performance due

to class rarity [5], since our approach shows an improvement in performance via a feature selection technique. We show that the feature selection technique improves the performance of One-Class GMM in the credit card fraud detection application domain. Moreover, the results we report for One-Class SVM show that some degree of robustness in models built with all features is necessary for SHAP feature selection to be effective. Two recent studies which show similar results of One-Class GMMs out-performing One-Class SVMs are in alignment with this finding [6], [7].

With the surging advancement of e-commerce, there has been a commensurate increase in the volume of credit card transactions. The scale of the volume is such that automated fraud detection is practical and cost-effective. Improving these systems requires high-volume, diverse transaction datasets. We focus on the Credit Card Fraud Detection Dataset [8] from Kaggle, an anonymized collection of financial transactions. Hereafter, we refer to the dataset as the “Credit Card” dataset. It is a valuable resource for the credit card fraud detection application domain. The dataset contains 284,807 financial transactions, and only 492 (0.172%) of the records are labeled fraudulent. We opted for this dataset owing to its authentic transaction data. Moreover, the dataset is a standard reference point for researchers to relate the performance of fraud detection systems. The extreme class imbalance makes this dataset well-suited for one-class classifiers that focus on modeling normal behavior to detect anomalies. We were unable to find a similar publicly available dataset with all of the qualities of the Credit Card dataset, hence we decided to use this one dataset for our study.

A challenge one may face with OCCs is determining feature importance for feature selection. For example, it would not be appropriate to use feature selection techniques applied by Wang *et al.* with OCCs, since these techniques involve Binary-Class Classifiers [9]. One-Class SVM and One-Class GMM are popular One-Class classifiers. On one hand, it is not possible to easily derive feature importance from the parameters that One-Class SVM and One-Class GMM learn during their training phases. On the other hand, it is possible to use SHAP to determine the feature importance of trained One-Class SVM and One-Class GMM models. Therefore, we employ SHAP in experiments with One-Class SVM and One-

Class GMM and the Credit Card dataset to discover which attributes of the dataset have the strongest influence in their classification outcomes in terms of Area Under the Precision Recall Curve (AUPRC) [10]. Here, we chose to focus on AUPRC since previous research has shown that the other well-known threshold-agnostic classification metric, Area Under the Receiver Operating Characteristic Curve (AUC), is a misleading metric for the classification of highly imbalanced data [11]. Once we know feature importance, we can do feature selection experiments where we limit the features used to build models on different sized subsets of the most important features. During our review of related work, we found a lack of research where SHAP is used to do such feature selection for OCC models.

Therefore, our work aims to bridge this gap by analyzing the application of SHAP to select the most informative features for OCCs. The results demonstrate that SHAP can identify a reduced set of features that can improve model accuracy while ignoring noisy or less useful inputs. This illuminates how we can interpret OCCs and enhance their effectiveness through feature selection. Our study provides evidence for the utility of SHAP for feature importance tasks with OCCs absent in prior work. This paper is organized into the following sections: Related Work, Algorithms (where we describe SHAP, One-Class GMM and One-Class SVM), Methodology, Results, Statistical Analysis, and Conclusions.

RELATED WORKS

Due to our interest in SHAP for feature selection in experiments involving OCCs, we conducted a literature review of studies that mention OCCs and SHAP. While we are able to find a limited number of such studies, we discovered that none of the studies leverage SHAP as a feature selection technique in conjunction with one-class classifiers. Moreover, we did not find studies which compare results of One-Class GMM, One-Class SVM and SHAP. Therefore, our study makes a novel contribution to the field of Machine Learning. For a broad survey

The first study we found, by Abadía *et al.* [12], uses One-Class SVM and SHAP as part of a framework that takes healthcare data from sensor networks as input for a One-Class SVM. However, the authors' motivation for employing SHAP is to explain which attributes of the input dataset have the largest effect on the behavior of a trained One-Class SVM. We have a different motivation for employing SHAP with OCCs. We use SHAP to determine feature importance. Then, we build multiple models with different numbers of features in experiments to determine whether OCCs yield similar or better performance with a smaller number of features. Therefore, our study makes a more extensive use of SHAP than Abadía *et al.*'s.

The second study we found that involves SHAP and OCCs is by Oki *et al.* [13] Their study involves the use of OCCs for anomaly detection in maritime equipment. Similar to Abadía *et al.*, Oki *et al.* propose the use of One-Class SVM for anomaly detection in sensor data, with the application

of SHAP to determine which sensor data have the biggest influence on anomaly detection. We did not find an indication that Oki *et al.* use SHAP for feature selection. Moreover, their study also does not involve a comparison of OCCs, whereas we present findings for both One-Class SVM, and One-Class GMM.

Research by Hoang *et al.* [14] is the third and final study we were able to locate involving SHAP and One-Class SVM. However, Hoang *et al.* also do not employ SHAP for feature selection, as we do. Similar to the related works we cite above, the authors use SHAP to determine feature importance for their model. They propose a new model for detecting anomalies in industrial control system sensor data. Their new model is a hybrid of several Machine Learning algorithms that includes One-Class SVM. They use SHAP for feature importance to explain their hybrid model. Their use of SHAP highlights its ability to help explain complex models. In addition to the fact that Hoang *et al.* do not employ SHAP as a means of feature selection, our study describes the novel employment of SHAP for feature selection with two commonly used, well-known OCCs, One-Class GMM and One-Class SVM, as opposed to an application of SHAP to an experimental hybrid model.

The related works that we found employ SHAP with One-Class SVM for feature importance. We did not find a study where SHAP is used to determine feature importance in One-Class GMM. Furthermore, we are unable to find any study where SHAP is used for feature selection. This highlights another novelty of our study, which is to evaluate the effect of SHAP-based feature selection on two OCCs. Therefore, our study makes a unique contribution that proves the efficacy of SHAP as a tool for feature selection in conjunction with One-Class classifiers.

ALGORITHMS

Here we provide background on the algorithms that are essential to this study. These are the algorithms for SHAP, One-Class SVM and One-Class GMM. For a broad study of algorithms available for one-class classification, please see [15].

SHAP

Lundberg and Lee introduce SHAP in their 2017 seminal paper "A unified approach to interpreting model predictions" [4]. Their motivation for proposing SHAP is to eliminate a trade-off between simple models, that are easy to explain, but not very accurate, and more complex models, that are more accurate, but more difficult to explain. SHAP uses a technique from Game Theory for determining the contribution that a player makes to the output of a coalition [16]. Lundberg and Lee apply Shapley's technique to calculate feature importance. They show that several previous techniques for calculating feature importance are instances of a class, and that their technique is a member of the same class.

The class is characterized by techniques that build ancillary models to calculate feature importance for specific input values of the primary model. In addition, their proposed technique

has three desired properties: local accuracy, missingness, and consistency. Local accuracy means that the ancillary model for calculating feature importance has an output value equal to the primary model's output value for the same point. Missingness is a requirement that if a feature is missing in the original dataset, then it should not contribute to the ancillary model's approximation of the primary model. Consistency refers to the idea that if we are comparing ancillary models for approximating different related primary models, and a feature has the same or larger effect on the primary models, then it should have the same effect on the ancillary models as well. Lundberg and Lee prove that their proposed technique is the unique instance of the class of feature importance techniques that has the local accuracy, missingness, and consistency properties. The theoretical justification that Lundberg and Lee provide for SHAP makes a compelling argument for practitioners to adopt it, since it makes some guarantees about providing a well-behaved method for approximating a model, which can then be used for determining feature importance.

In their seminal paper on SHAP, Lundberg and Lee propose several variations of SHAP. Here, we use the first formulation of SHAP that Lee and Lundberg provide, Kernel SHAP. Kernel SHAP does not rely on the class of the primary model that it approximates, so it is applicable to any model. In other words, Kernel SHAP is model-agnostic. In their first paper on SHAP, Lundberg and Lee also cover more efficient techniques to calculate the SHAP feature importance that exploit properties of specific types of models. Here, we leverage the original SHAP values for feature importance that Lundberg and Lee propose in their seminal paper. SHAP values for a feature can be negative or positive real numbers, and their absolute values indicate the feature importance. Since SHAP values are calculated for an instance of a dataset, one may average SHAP values for several instances to get an aggregate notion of feature importance for a dataset.

SHAP is publicly available as an open source library for the Python programming language [17]. In addition to the SHAP values for feature importance, this library also supplies several tools for visualizing SHAP feature importance values. Examples are plots that visualize attribute contributions to model output, plots that visualize the effect of changes in model inputs to changes in model outputs, and local importance plots, which show the contributions of attributes of a particular instance to model output. Please see the provided reference for details on utilizing these visualizations. The key take-away practitioners should know about SHAP is that it is a readily available, model-agnostic technique for providing feature importance with theoretical guarantees on its feature importance calculations.

One-Class SVM

The scikit-learn implementation of One-Class SVM we use functions as described by Schölkopf *et al.* [18]. One may view One-Class SVM as a specialization of SVM which can be trained on data where all instances have the same label. The key concept of SVM is the maximum margin-

separating hyperplane. In a binary classification setting, this is the boundary that is maximally distant from the borders of two regions containing instances of either class. SVM fits a function to approximate the maximum margin-separating hyperplane of the training sample of a dataset. Classification of test samples is performed by assigning an instance to the same class as the other instances that lie of the same side of the maximum margin-separating hyperplane. Different kernel functions may be employed with SVM to project the features of a dataset into a space where its points are separable by a hyperplane. One-Class SVM does not have two classes of points in the feature space of a dataset. Therefore, it finds the maximum margin-separating hyperplane between its training data and the origin. The origin is the point in the feature space that has the value zero in all dimensions. The scikit-learn implementation of SVM OCC we utilize here allows one to select a radial basis function (RBF) [19] kernel. The use of this kernel transforms the procedure for calculating the maximum margin-separating hyperplane to one for finding a surrounding hypersphere that contains most of the instances of the training data. One-Class SVM with an RBF kernel can then be conceptualized as classifying instances of the test data according to whether the instance is contained in a hypersphere of the feature space after it is transformed by the radial basis function.

One-Class GMM

We use a scikit-learn implementation of One-Class GMM which leverages the expectation-maximization algorithm to fit a collection of Gaussian distributions to data for classification [20]. One-Class GMMs model the training data as a mixture, or sum of, multivariate Gaussian distributions. Each Gaussian component may cover separate partitions of instances in the training data. During training, One-Class GMMs fit a number of Gaussian components to the target data, which consists of only instances of one class. The process of fitting a One-Class GMM entails learning the mean and covariance parameters that best describe the data. After training on instances of a single class, to perform a classification, One-Class GMM computes the likelihood that a new test instance was generated by the distributions it fit to the training data. If the calculated likelihood drops below a certain level, the system identifies the instance as a member of the class not used for training.

Calibration

One-Class SVM is a popular algorithm for anomaly detection, but the scikit-learn implementation we use does not provide probability estimates. The AUPRC metric we use to evaluate classification performance requires class probability estimates as inputs. To counter this, we employed sigmoid calibration [21]. Sigmoid calibration involves converting the output of the One-Class SVM algorithm using a logistic function. This enables us to convert the One-Class SVM output scores into class probabilities.

We found sigmoid calibration to be helpful in calculating AUPRC for One-Class GMM as well. We utilized the

GaussianMixture model from scikit-learn version 1.2.0 for our Gaussian Mixture Model One-Class GMM implementation. In preliminary experiments, we found the `predict_proba` method of this One-Class GMM version yielded class membership probabilities concentrated near 0.0 or 1.0. Calculating precision and recall for these probability outputs is unchanged across most threshold values for plotting the precision-recall curve. The limited set of precision and recall scores results in a precision-recall curve lacking enough points to be meaningful. We concluded the `predict_proba` function of `sklearn.mixture.GaussianMixture` lacks robustness. Therefore, we applied sigmoid calibration to convert the GMM outputs into well-calibrated probabilities [22], [23]. Calibration enabled a more finely spaced, interpretable precision-recall curve.

METHODOLOGY

Our method for conducting experiments is a process with two phases. Each phase is implemented with the execution of a Python program. The first phase is feature selection, and the second is model evaluation.

In both phases we work with the Credit Card dataset mentioned in the introduction. The Credit Card dataset is derived from European credit card transaction data from September 2013. It has 30 features, and one label. There is one feature, “time”, which we do not use since it would act as a unique identifier and models could use it to memorize the labels of the training data, and fail to generalize. To preserve the privacy of individuals involved in the transactions that constitute the Credit Card data, transaction data was transformed via principal component analysis (PCA) [24], except for the “time” and “transaction amount” features. The features that are the output of PCA are labeled V1, V2, ... V28. Features are scaled to the [0, 1] range before they are input to the classifiers for training or inference.

We use default values for the One-Class GMM classifier’s hyperparameters, except for the `n_components` parameter, which we set to 2. For One-Class SVM we set the `nu` parameter to 0.0025, the `kernel` parameter to “rbf”, and the `gamma` parameter to 0.0310. We use the same hyperparameter values for classifiers during SHAP feature selection, and classification experiments. Hyperparameter values were selected with a grid search.

For feature selection, we employ the Python implementation of SHAP, which is a library with the same name as SHAP, but spelled in lower case letters. Therefore, when we refer to SHAP (in upper case letters), we refer to the theoretical concepts that Lundberg and Lee use to define SHAP. When we refer to shap (in lower case letters), we refer to the implementation of the SHAP concepts in the Python programming language. We use shap version 0.41.0 [17] for feature selection, and ten rounds of five-fold cross validation for model evaluation. Data is separated with scikit-learn’s `k-fold cross validation` library, where `k` is set to five. Since we work with one-class classifiers, after the data is separated into train and test subsets, instances of the minority class are discarded from

the training data. We do not alter the class distribution in the test partition of the dataset.

In the first phase, we train an uncalibrated One-Class SVM or One-Class GMM model on the entire dataset. We then pass the trained model and a sample of the dataset to the SHAP `KernelExplainer` object. We use the `shap.sample` function to take the sample of the dataset. We use `shap.sample`’s default value of 100 for the number of samples. After initializing the `KernelExplainer` object, we invoke its `shap_values` function, which yields a sorted list of the Credit Card dataset’s features. The `shap_values` function returns a two-dimensional array of SHAP (feature importance) values for every sample. We take the mean absolute value of each row of SHAP values to arrive at a feature importance value for each feature. In the documentation for the shap library, Lundberg, co-author of the seminal paper on SHAP, mentions that he uses the same technique of averaging the absolute values of the SHAP scores of features.¹ We then sort the list of features by their mean SHAP values. The lists in Table I are the result of the technique described here, truncated at the feature with the 15th highest SHAP value.

In the second phase, we leverage the feature importance lists we created during the first phase. We build One-Class SVM and One-Class GMM models with the n features of highest SHAP feature importance, where n takes the values three, five, seven, ten, and 15. For validation of the effectiveness of feature selection, we also build models with all 29 features of the Credit Card data. For One-Class GMM we use the feature importance lists that we generated with GMM and SHAP in phase one, and for One-Class SVM we use the feature importance lists we generated with SVM and SHAP in phase one. We run ten iterations of five-fold cross validation for each model. For each test fold, we calculate and record the AUPRC score the model yields.

AUPRC serves as a metric to gauge the trade-off between precision and recall. Precision is the fraction of true positives that are correctly identified, while recall is the fraction of all positives that are correctly identified. AUPRC is calculated by plotting precision against recall for different classification thresholds, and then taking the area under the curve. A higher AUPRC value indicates better performance. AUPRC has a maximum value of one, and a minimum value of zero. Previous research also shows that AUPRC is a better metric for classification of the Credit card data [25].

Since we execute ten iterations of five-fold cross validation, we record a total of 50 AUPRC scores for each model. Therefore, for each classifier, because we have six possible values for the number of features to use, we have a total of $6 \times 50 = 300$ AUPRC scores.

We execute all of our experiments on CPUs in a high performance computing environment. The environment consists of two clusters of CPU nodes. In one cluster each node has an Intel Xeon processor with 164 GB RAM, and in the other cluster, each node has an AMD Ryzen processor with 1TB

¹<https://github.com/shap/shap/issues/632#issuecomment-503778940>

RAM. One node of either cluster has sufficient resources to execute the tasks we describe in phase one and phase two.

RESULTS AND DISCUSSION

Our first result from phase one is the feature importance lists we obtain by applying SHAP to the One-Class SVM and One-Class GMM models. Table I contains the top 15 most important features by classifier, according to their SHAP scores. Please see the Methodology Section for a discussion of how we convert SHAP scores for features to a ranking. Since SHAP uses the trained model to calculate feature importance, it yields a different ranking of features for each model.

TABLE I
TOP 15 FEATURES SELECTED BY THEIR SHAP FEATURE IMPORTANCE VALUES; THE FEATURES ARE LISTED IN ORDER OF THEIR SHAP FEATURE IMPORTANCE VALUES FROM LEFT TO RIGHT; FEATURES PRESENT IN BOTH LISTS ARE IN BOLD TEXT

SVM:	V1	V23	V25	V3	V18	V28	V2	V19
	V21	V15	V5	V20	V4	V11	V16	
GMM:	V11	V2	V26	V12	V8	V5	V9	V7
	V18	V3	V10	V16	V17	V21	V14	

Table II contains the mean AUPRC scores for the One-Class GMM. Each AUPRC score reported is the mean of 50 experimental outcomes. Table II shows a positive trend in AUPRC scores with up to 15 features. When all features are used, One-Class GMM yields a lower AUPRC score, indicating that our feature selection technique by SHAP values improves One-Class GMM's fraud detection capability.

TABLE II
MEAN ONE-CLASS GMM AUPRC SCORES BY THE NUMBER OF FEATURES SELECTED

Features	AUPRC
3	0.2515
5	0.3669
7	0.3411
10	0.4071
15	0.6420
29	0.4971

Table III holds the mean AUPRC scores for One-Class SVM. We see that SVM's performance improves as we add more features. However, unlike One-Class GMM, SVM's performance continues to improve in the case where we use all features. This implies that, unlike GMM, SHAP does not provide a benefit to SVM in terms of AUPRC scores. We conjecture that this is due to the initial performance of SVM, which is not strong enough to provide SHAP with enough information to effectively select features.

TABLE III
MEAN ONE-CLASS SVM AUPRC SCORES BY THE NUMBER OF FEATURES SELECTED

Features	AUPRC
3	0.0114
5	0.1748
7	0.1644
10	0.1429
15	0.2835
29	0.3775

STATISTICAL ANALYSIS

We conduct a statistical analysis of the experimental outcomes summarized in Tables II and III. The general method of our statistical analysis is to conduct an analysis of variance (ANOVA) [26] test where we treat the number of features as a factor in the experiments. When the $\Pr(>F)$, or p -value in the result of the ANOVA test is less than or equal to our pre-selected threshold (significance level) of 0.01, we conclude that the number of features has a significant effect on the experimental outcomes. When such a conclusion is possible we conduct a Tukey's Honestly Significant Difference (HSD) [27] test to rank the levels of experimental factors in terms of their impact on experimental outcomes. For the purpose of our study, we define experimental outcomes as the AUPRC scores. Therefore, the ANOVA test results inform us whether the number of features has a significant effect on the AUPRC scores that the One-Class SVM and One-Class GMM models yield.

The outcome of the HSD test is a ranking of the experimental factors according to their effect on experimental outcomes. The factors with rank 'a' are associated with the highest AUPRC scores, those with rank 'b' are associated with the second highest AUPRC scores, and so on.

One factor ANOVA for GMM Experiments with Credit Card Datasets, Analysis of Results in Terms of AUPRC

Table IV holds the ANOVA test result for experiments with One-Class GMM and SHAP, where the number of features is treated as an experimental factor. Since the p -value we calculate as the result of the test is practically zero, the HSD test is applicable.

TABLE IV
ANOVA FOR FEATURES AS A FACTOR OF ONE CLASS GMM'S PERFORMANCE IN TERMS OF AUPRC

	Df	Sum Sq	Mean Sq	F value	$\Pr(>F)$
Features	5	4.64	0.93	264.16	*
Residuals	294	1.03	0.00		

* indicates the value is less than 1×10^{-4}

Table V contains the HSD test results for the number of features factor in the GMM SHAP experiments. The HSD test result confirms the trend we mention regarding the AUPRC scores we present for One-Class GMM in Table II. One-Class GMM models built with the 15 most important features yield

significantly better AUPRC scores than models built with all features.

TABLE V
HSD TEST GROUPINGS AFTER ANOVA OF ONE CLASS-GMM AUPRC
SCORES FOR THE FEATURES FACTOR

Group a consists of: 15
Group b consists of: 29
Group c consists of: 10
Group cd consists of: 5
Group d consists of: 7
Group e consists of: 3

One factor ANOVA for SVM Experiments with Credit Card Datasets, Analysis of Results in Terms of AUPRC

We present the outcome of the ANOVA test for the effect of feature selection on SVM SHAP's AUPRC scores in Table VI. Similar to the results for One-Class GMM, we find that the number of features used to train One-Class SVM has a significant effect on the experimental outcomes. Therefore, an HSD test enables us to rank the number of features factor and determine which number of features yields the best performance.

TABLE VI
ANOVA FOR FEATURES AS A FACTOR OF ONE-CLASS SVM'S
PERFORMANCE IN TERMS OF AUPRC

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Features	5	3.94	0.79	718.36	*
Residuals	294	0.32	0.00		

* indicates the value is less than 1×10^{-4}

The HSD test result in Table VII confirms the conclusion hinted at by the AUPRC scores listed for One-Class SVM in Table III. Although the number of features factor has a significant effect on experimental outcomes, we find that the use of all features is associated with the best performance. This implies that the SHAP feature selection technique is not useful, since One-Class SVM yields better AUPRC scores without it. As mentioned previously, we conjecture this is due to the initial performance of One-Class SVM with all features of the Credit Card dataset.

TABLE VII
HSD TEST GROUPINGS AFTER ANOVA OF ONE-CLASS SVM AUPRC
SCORES FOR THE FEATURES FACTOR

Group a consists of: 29
Group b consists of: 15
Group c consists of: 5
Group cd consists of: 7
Group d consists of: 10
Group e consists of: 3

CONCLUSIONS

The results of our experiments show that our novel technique of selecting features based on SHAP feature importance for OCCs is effective. We make this conclusion on the basis of the performance of One-Class GMM with features selected according to SHAP feature importance. One-Class GMM models built with the 15 most important features according to SHAP show significantly better performance than One-Class GMM models built with all features of the Credit Card dataset.

A second conclusion we draw from our results is that some models benefit from SHAP feature selection, while others do not. We suspect that the initial, worse performance of One-Class SVM impacts the quality of the feature ranking that SHAP provides. Put another way, since One-Class SVM does not yield strong performance to begin with, it is not a good source of information about which features are useful for the credit card fraud detection machine learning task. The subject of future research will be to apply the SHAP feature selection technique to other application domains where one-class classifiers are an appropriate choice.

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